

Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

Hema Gujjar

hema.hgujjar@gmail.com

Katherine MacKay

kmacusa2017@gmail.com

Gaelle Ngabo

gaelle.ngabo@gmail.com

Parker Sherwood

parker.sherwood@icloud.com

Samuel Tobachnick

stobach09@gmail.com

Krishiv Patel*

krishivp23@gmail.com

New Jersey Governor's School of Engineering and Technology

July 31, 2026

*Corresponding Author

Abstract—Airfoils are one of the most fundamental components of aviation and aerospace engineering, yet relatively little is known about the aerodynamic interactions between airfoils in biplane configurations, as most studies tend to solely focus on mono planes. This study addresses this gap by analyzing airplanes in both mono and biplane configurations. The study focuses on 4 different airfoils: NACA 0012, 2212, 2412, and 2612. For the initial theoretical simulation and analysis, 2D simulations of these airfoils were conducted in XFLR5 and ANSYS Fluent before experimental testing. Following the theoretical analysis, Models were 3D printed and tested in a wind tunnel at a Reynolds number of $Re = 20100$, and 51 biplane configurations were tested in combinations determined by airfoil shape and decalage test matrices. Furthermore, finite-span aspect-ratio corrections were applied to the computation fluid dynamic (CFD) results to distinguish 3D downwash effects from the behavior of low-Reynolds-number viscous flow. The experimental results demonstrated that aerodynamic efficiency, measured by lift-to-drag ratio (L/D), was influenced more through changes in lift rather than in drag. Of the airfoil shape combinations, the NACA 2212 x NACA 2212 combination produced the highest efficiency at baseline conditions. Following this finding, further testing was taken on this biplane. After further decalage testing, it was found that the biplane produced a maximum efficiency of 9.12 with the top airfoil at 0° angle of attack and the bottom angle at 5° angle of attack. The most efficient combinations generally used a converging wing combination that accelerated airflow through the gap, generating a concentrated venturi effect, while negative lower-wing angles caused noticeable losses in lift and an overall worse performance. These findings demonstrate the importance of airfoil selection and decalage in determining the aerodynamic efficiency of low-Reynolds-number biplane configurations.

I. INTRODUCTION

The idea of biplanes (two sets of wings stacked one above the other) has been considered and studied ever since the

inception of airplanes. Many of the first airplanes, such as the Wright Flyer, were biplanes. The main motivation behind early biplanes was to reduce weight and to provide structural support for the wings [1]. During the first half of the twentieth century, biplanes were extensively studied, but over time they lost popularity in favor of monoplanes (one-winged airplanes). Using Computational Fluid Dynamics (CFD) simulations, 2-dimensional airfoils were modeled and simulated to graph the streamlines and velocity contours around the airfoils, along with graphs displaying the relationship between the coefficients of lift and angle of attack, coefficients of drag and angle of attack, coefficient of lift and the coefficient of drag (referred to as aerodynamic efficiency), and the aerodynamic efficiency and the angle of attack. This CFD software can be used to model any shape, 2D or 3D, and make predictions about the behavior of various shapes, in our case airfoils. However, because the airfoils are 2D and the computers used are limited in computational power, there are many limitations on the accuracy of the simulations, which increases the importance of experimental tests. Using these graphs, informed decisions were made about the behavior of the selected airfoils to determine their suitability for biplane testing in the wind tunnel. After the simulations, the airfoils of interest were modeled using Computer-Aided Design (CAD) software and 3D-printed for testing in a wind tunnel at subsonic speeds. The motivation to focus on biplanes, rather than monoplanes, stemmed from the limited modern research in the field, particularly the use of experimentally derived data. The decision was also influenced by the large current body of research and experimentation on monoplane airfoils, especially NACA airfoils. Biplanes are useful alternatives to monoplanes in many applications due

to their distinct behavior and structure, such as crop dusting and landing on short airfields, partly due to their greater maneuverability, shorter wingspans, and higher lift at low speeds [2].

II. BACKGROUND

A. Airfoils

An airfoil is a two-dimensional cross-section of a wing [3]. They are among the most essential components of aerospace engineering. Most airfoils have the express goal of generating enough lift to overcome the force of gravity on the aircraft. Airfoils are constructed around three primary measurements: the leading and trailing edges, the chord line, and the camber line.

The leading (LE) and trailing (TE) edges are the front and rearmost edges of the airfoil. These points are connected by the chord, a straight line representing the overall length of the airfoil. The chord (c) serves as an essential reference for a wide variety of related calculations pertaining to the geometry and aerodynamics of the airfoil. The LE and TE are also connected by the camber line, which determines the airfoil's curved geometry. Depending on the configuration of these measurements, an airfoil's geometry can be tailored for specific situations[3].

In the late 1920s and early 1930s, the National Advisory Committee for Aeronautics (NACA) created a system of four-digit codes to identify airfoils based on several metrics [4]. The first metric addressed in a NACA code is the maximum camber (m), expressed as a percentage of the chord length.

$$m = c \frac{M}{100} \quad (1)$$

The max camber describes the distance between the chord line and the highest point on the camber line. Following max camber in the NACA code is camber location (p), which is communicated by a fraction of ten on the chord line [4].

$$p = c \frac{P}{10} \quad (2)$$

Camber location, coupled with max camber, describes the average curvature of the airfoil. While max camber describes the asymmetry of the airfoil, the camber location dictates where this effect occurs. The final two digits of the NACA code represent the max thickness (t) of the airfoil as a percentage of the chord [3].

$$t = c \frac{XX}{100} \quad (3)$$

Unlike camber, max thickness does not require a digit describing its location, as the location of this measurement is fixed by default at 30% of the chord ($0.3c$) from the leading edge for 4-digit NACA airfoils.

Symmetric airfoils can be used to illustrate the importance of each respective NACA number. In a symmetric airfoil, the camber line is completely horizontal, lying along the chord line [3]. Consequently, a symmetric airfoil's NACA number

will begin with two zeros, as there is no asymmetry in the geometry for the max camber to describe and therefore no location of the max camber. Only max thickness is required to describe the geometry of a symmetric airfoil.

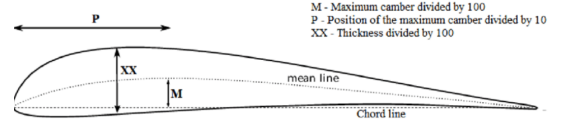


Fig. 1: Illustration of core airfoil measurements

B. Aerodynamic Coefficients and Parameters

1) *Angle of Attack*: Angle of Attack (α) is defined as the angle formed between a wing's chord line and the oncoming relative wind (U_∞) [See Fig. 2] [5]. The angle of attack directly controls the coefficients of lift, drag, and the stalling point¹ of an aircraft. By changing the angle of attack, airflow is redirected around the wing, creating a velocity difference between the upper and lower surfaces. This difference in airflow ultimately generates lift [6].

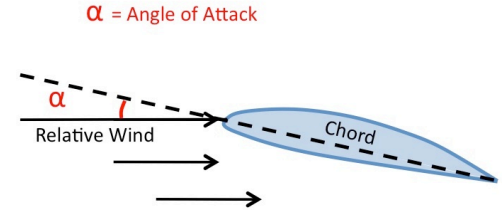


Fig. 2: Diagram of angle of attack with respect to wind and airfoil.

As angle of attack increases, air passing over the upper surface accelerates much faster than the air along the bottom surface ($V_{top} > V_{bottom}$)². Based on Bernoulli's Principle:

$$P + \frac{1}{2}\rho U^2 + \rho gh = constant \quad (4)$$

$$P_1 + \frac{1}{2}\rho U_1^2 = P_2 + \frac{1}{2}\rho U_2^2 \quad (5)$$

With P_1 = static pressure on the upper surface, U_1 = airflow velocity over the upper surface, P_2 = static pressure on the lower surface, U_2 = airflow velocity under the lower surface, and ρ = density of the air.

The velocity gap between V_{top} and V_{bottom} generates higher pressure beneath the airfoil ($P_{bottom} > P_{top}$), creating a static pressure difference (ΔP):

¹See section Flow separation and Stalling

² V is used for local surface velocities, air directly along the upper and lower surfaces of the airfoil. U_∞ represents freestream velocity, or air far away from the wing [3]

$$\Delta P = P_2 - P_1 = \frac{1}{2}\rho(U_2^2 - U_1^2) \quad (6)$$

Given by rearranging the terms in the above equation, this pushes the wing upward. Symmetric airfoils (such as NACA 0012) have identical curves on the top and the bottom, meaning they do not generate any lift at a zero angle of attack ($C_L = 0$ at $\alpha = 0^\circ$). Cambered airfoils, however, are naturally asymmetric, meaning they can produce lift at zero angle of attack [4]. To reach zero lift, a cambered airfoil must be tilted to a negative angle. In the following experiment, testing different values of α allows us to analyze how changes in camber location and height affect airfoil performance across varying α values. Furthermore, mapping these performance trends across the full angle-of-attack range provides essential baseline data for comparing experimental wind-tunnel metrics with computational predictions from XFLR5 and ANSYS. This comparison ultimately highlights how variations in geometric profile influence boundary layer stability and aerodynamic efficiency across different operational regimes.

2) *Coefficient of Lift and Drag:* To compare airfoils regardless of their scale, forces can be converted into dimensionless coefficients [3]. Coefficient of Lift (C_L) and Coefficient of Drag (C_D) are created using dynamic pressure (q), planform area (A), and the lift and drag force (L and D , respectively). The following are equations for each of these terms.

$$q = \frac{1}{2}\rho U^2 \quad (7)$$

$$A = cb \quad (8)$$

$$C_L = \frac{L}{\frac{1}{2}\rho U^2 A} = \frac{L}{qA} \quad (9)$$

$$C_D = \frac{D}{\frac{1}{2}\rho U^2 A} = \frac{D}{qA} \quad (10)$$

Planform area describes the area of the entire wing if looking at the aircraft from a top-down view; in our case, the wing has a rectangular planform area, meaning the wing forms the shape of a rectangle, so we can calculate A by simply multiplying the chord (c) and the span (b , the length of the wing from wingtip to wingtip) [7]. For the purposes of our experiment, we will define the Lift force as the force normal to the relative wind, and the drag force as the force parallel to it. A positive lift force would be directed upwards relative to the object's standard orientation, whereas a negative lift force would be forcing the aircraft downward. A positive drag force would be acting in the same direction as the relative wind in the exact opposite direction of the relative motion of the aircraft.

C_L measures how efficiently the shape of the airfoil generates lift, while C_D measures resistance. In normal operating conditions, C_L increases linearly as α increases.

3) *Stagnation Points:* When air hits the airfoil, it splits at two stagnation points. Flow velocity drops to 0 at that point and moves either up or down and around the wing, converting kinetic energy into maximum static pressure. [See Fig. 3]

Stagnation points are not to be confused with the leading edge and trailing edge of the airfoil. While the stagnation points are the same as the LE and TE on a symmetric airfoil, adding an angle of attack shifts them depending on the value of α . For the leading edge stagnation point (S_1), as the angle of attack increases, S_1 shifts downward across the bottom of the wing. When the angle of attack decreases, S_1 starts to shift upward across the top of the wing. The trailing edge stagnation point (S_2), however, always remains on the TE and is the point where the air flowing on the top and bottom of the wing rejoins [4].

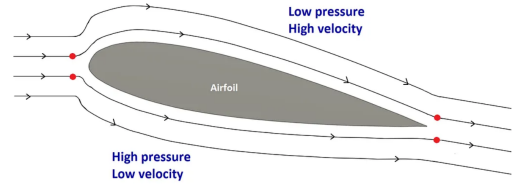


Fig. 3: Diagram showing airflow with respect to stagnation points

As S_1 shifts further down the lower surface under higher angles of attack, air destined for the upper surface is forced to negotiate a sharper turn around the leading-edge radius. This rapid local acceleration establishes a strong localized low-pressure zone known as the suction peak. Understanding the behavior and movement of these stagnation points is critical for predicting where boundary layer separation begins, as the location and intensity of the suction peak directly dictate the strength of the downstream adverse pressure gradient leading to stall [3].

4) *Flow separation and stalling:* At lower angles of attack, air hugs around the airfoil cleanly. However, as air passes the suction peak³ and moves towards the trailing edge, it has to travel to the back of the wing, where the pressure is much higher.

Moving from low to high pressure is called an adverse pressure gradient. This means the air is forced to push uphill against the rising pressure. The air inside the boundary layer⁴ has already been slowed by friction on the wing surface. If the air does not have enough kinetic energy to resist the rising pressure, it slows down, stops, reverses direction, and is pushed away from the wing. This detachment is called flow separation [3].

Regardless of the size, speed, or span, each airfoil has its specific critical angle of attack (α_{crit}). If an airfoil has an angle

³The suction peak is the exact spot on the top surface of the wing where air moves the fastest, and the air pressure drops to its lowest point.

⁴The boundary layer on an airfoil is the very thin layer of air right next to the surface of the wing where friction slows the air down.

of attack that exceeds its α_{crit} , the air can no longer smoothly follow the upper surface of the wing. The airflow separates, creating turbulence and interfering with the pressure difference that ultimately generates lift. This sudden loss in lift is called aerodynamic stall. [See Fig. 4]

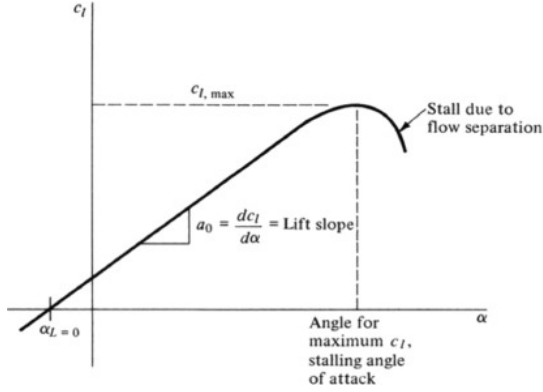


Fig. 4: Diagram demonstrating relationship between C_L and α with the stalling point

5) *Laminar vs Turbulent Air, Reynolds Number:* Airflow is primarily categorized into two main fluid-dynamic patterns: laminar flow and turbulent flow.

Laminar air moves in smooth, parallel layers without intersections or mixing. This type of air moves at a constant velocity either vertically or horizontally. This type of air is optimal for aircraft flight because it flows smoothly over the aircraft's surface, creating minimal friction and drag.

Turbulent air, on the other hand, is characterized by its unsteady mixtures of air flow with irregular paths and strong inertial forces. That being said, while this rough and chaotic mix of air particles increases drag, it helps delay aerodynamic stall by keeping the air attached to the wing at high angles of attack [8]. [See Fig. 5]

The Reynolds number is a dimensionless value that separates air flow into laminar and turbulent states based on the ratio of internal forces and viscous forces. The Reynolds number (Re) is defined as:

$$Re = \frac{\rho v L}{\mu} \quad (11)$$

where:

- ρ is the density of the air,
- v is the velocity of the airflow,
- L is the characteristic length (e.g., chord length),
- μ is the dynamic viscosity of air.

A Reynolds number of less than 2300 means laminar flow. Greater than 4000 means turbulent flow, and between the two signifies transitional flow, which is a mixture of both laminar and turbulent air flows.

C. Biplanes

The information outlined in the following section was sourced from Max M. Munk's paper *General Biplane Theory* [2]

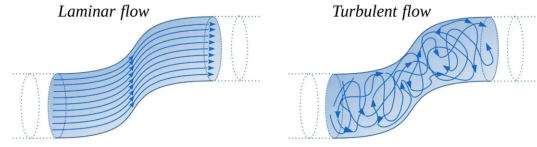


Fig. 5: Arrows demonstrating both laminar and turbulent flow

1) *General Background:* Along with analyzing the aerodynamics experimentally and computationally of single airfoils, analysis was performed on stacked airfoils of several metrics to simulate the performance of biplanes. Biplanes are a type of aircraft in which two wings are stacked vertically on top of each other. This results in a seemingly larger lift generating surface in a shorter span without resulting in a flimsy, unstable structure. The term gap refers to the vertical distance between the upper-wing chord line and lower-wing chord line. Similarly, the term bay refers to the area between the struts connecting the upper and lower wing. The horizontal fore-aft positioning of the upper wing relative to the lower wing is known as stagger. In other words, it is the forward or backward displacement of the top wing with respect to the lower. A positive stagger is when the top wing is displaced ahead of the lower wing, and the opposite direction for negative stagger. In respect to aerodynamics, positive stagger is generally more efficient as it tends to reduce the center of drag whereby stability is improved [2]. Biplanes are generally considered inefficient due to several aerodynamic factors. Specifically, the gap tends to be small enough to cause airflow interference which also drastically reduces net lift despite having a larger lift generating surface. Additionally the struts between the wings also add drag which is amplified at higher speeds. One crucial aspect of improving aerodynamic efficiency is controlling wing tip vortices. Winglets, vertical or angled extensions placed at wing tips, can be used to spread out wing tip vortices. Additionally, biplanes are another promising possibility to consider in the aspect of controlling wingtip vortices. Apart from this, configuring light weight, but strong structures tends to be easier for biplanes than monoplanes. Currently, biplanes are not optimally considered for most aircraft applications, but for scenarios with restricted wing spans and lower operational speeds, they provide a considerable advantage [2]. Due to being deemed inferior to monoplanes, minimal extensive research has been performed on biplanes. Specifically in terms of analysis of factors such as stagger, angled positioning, and different airfoils on the upper and lower wings. A computational and experimental analysis and optimization of these factors has potential to reveal new and novel insights significant to the field of aerospace engineering as a whole [9]. The novelty is not just a source of the possible applications, but bridging a gap in the lack of understanding in biplane aerodynamics which can provide deeper understanding of countless physical phenomena and aspects crucial to aerospace engineering. This specific study focuses on analyzing different airfoils configurations for upper

and lower wings followed by decalage optimization for the best performing configuration.

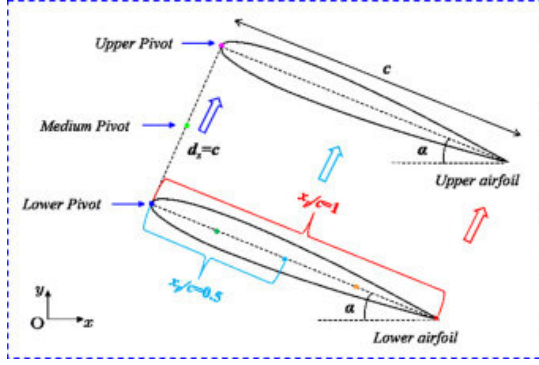


Fig. 6: Biplane Diagram

D. Theoretical Modeling of Biplane Airfoils

The length of the wing at a right angle to the chord is generally small compared to the entire chord and gap. However, it cannot be assumed that the chord itself is small compared to the gap. Due to this, the wings cannot be treated as points, but since the wing's thickness is small enough compared to the chord and gap, they can be simplified as two finite lines. Essentially, Munk starts by solving for the airflow around two line segments and later generalizing them to real shapes.

However, Munk runs into an issue when treating the wings as two line segments [2]. The angle of attack and speed give infinitely many flow possibilities depending on the amount of circulation added in. In reality, only one of these flow patterns happens. In order to draw the straight line estimation at the correct angle relative to the real cambered airfoil, Munk picks the angle such that the straight line experiences the same moment about the wing's center as the actual airfoil does at the zero-lift angle of attack. Through this calculation, he observes that this moment-matching angle ends up being very close to the chord line itself (off by about 2 degrees):

$$\alpha = \frac{2}{\pi} C_{m,0} \quad (12)$$

Still after this, he notes that a straight line at zero angle of attack still has zero lift, but a real cambered airfoil will have lift even at zero angle due to curvature itself. To solve this, Munk imagined the flow around the cambered airfoil that would have the least possible kinetic energy, and called this the section flow. Least kinetic energy refers to the flow with zero circulation, which results in a velocity integral equal to zero as it is around a closed path, so zero lift is produced.⁵ Physically, however, since there is no circulation, the theoretical velocity at the trailing edge is infinite. Munk names this edge velocity and describes it with the equation:

$$v_{\text{edge}} = \frac{m}{\sqrt{e}} \quad (13)$$

⁵By the Kutta Joukowski theorem, zero circulation yields zero lift [10].

where e is a small distance from the edge and m is a constant.

In reality, nature doesn't produce a pure section flow since infinite velocity is not possible. Instead, the circulation flow is superposed on top of it. It also has infinite velocity at the trailing edge, but of opposite sign so it cancels with the other. Essentially, it is the amount of circulation that makes the flow smooth rather than infinite. Munk then maps the same edge velocity that the cambered airfoils section flow has to the straight line. Now, the lift and moment match the real airfoils airflow behavior. Therefore, the influence between two such lines correctly replicates the aerodynamics of two cambered airfoils.



Fig. 7: Section flow without circulation

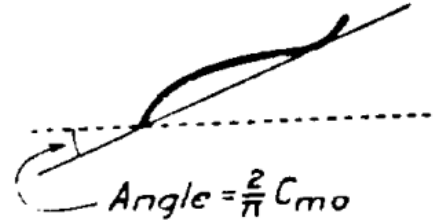


Fig. 8: Moment-matching angle for cambered airfoils

1) *Unstaggered Equal-Wing Biplanes*: First off, as a baseline, Munk calculated the magnitude of the edge velocity for a single cambered wing. To begin the calculation, Munk began with the key measurements pertaining to the airfoil. In his calculations, z represents a position along the wing with $z = 0$ at the center of the wing, $\pm T/2$ are the tips (leading and trailing edges), and T defines the chord length. The curvature of the wing is represented by:

$$\beta_0 = \frac{1}{2} \pi C_{l,0} \quad (14)$$

where $C_{l,0}$ is the lift coefficient when the real moment is zero as mentioned in the previous section, and β_0 converts the lift into the equivalent angle. Finally, the velocity of the vertical flow (responsible for angle-of-attack lift) is:

$$\frac{dW}{dz} = iV \sin \beta_0 \frac{z}{\sqrt{(T/2)^2 - z^2}} \quad (15)$$

Munk efficiently used complex numbers to help track both the speed and direction of flow simultaneously. In the equation, as z approaches $T/2$, the denominator approaches

zero, making the whole expression rapidly increase to infinity. Mathematically, this demonstrates the sharp edge velocity mentioned before. To quantify further, Munk defined e as the tiny distance from the trailing edge. Therefore, $z = T/2 + e$ clarifies the algebra near $e \rightarrow 0$. Employing this logic reveals the equation:

$$\frac{dW}{dz} = iV \sin \beta_0 \frac{\sqrt{T/2}}{\sqrt{2e}} \quad (16)$$

The strength of the infinity (the measure of the increasing, asymptotic behavior near the edge) equal to the constant m mentioned previously is taken from the equation above and is illustrated by:

$$m = iV \sin \beta_0 \frac{1}{2} \sqrt{T} \quad (17)$$

Following this, he considered two wings stacked with a gap G in between and chord T . Directly solving for the flow is incredibly difficult, so Munk used a type of coordinate transformation to efficiently solve the problem. He mapped the biplanes two horizontal wing-lines into two vertical line segments stacked on a single vertical line in an imaginary t -plane running from $t = 1$ to $t = k'$ and from $t = -1$ to $t = -k'$.

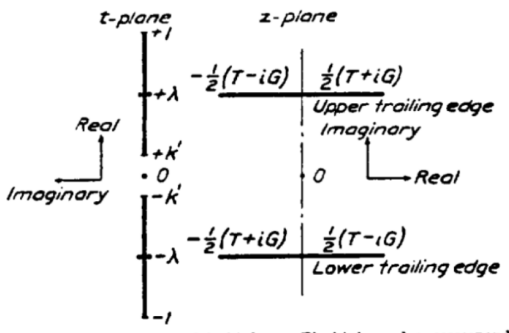


Fig. 9: Transformation of the biplane. The biplane edges correspond to the points.

From this new mapping, he derived:

$$\frac{dz}{dt} = C \frac{T}{2} \frac{t^2 - \lambda^2}{\sqrt{(1-t^2)(t^2 - k'^2)}} \quad (18)$$

Intuitively, it can be noted that C , λ , and k' are constants chosen specifically such that this mapping correctly reproduces the biplane's actual gap (G) and chord (T). The points $t = \pm \lambda$ correspond to the wingtips of the biplane wings. Additionally, G and T translate into computations involving complex elliptical integrals to compute the values. Furthermore, Munk worked out the same edge velocity calculation from before, but for the two-wing case. Munk solved for this velocity in the warped t -space and converted it back to the physical velocity using the same transformation:

$$B = \frac{1}{2} \frac{C}{\lambda} \sqrt{(1-\lambda^2)(\lambda^2 - k'^2)} \quad (19)$$

$$e = -\frac{T/2}{2B} \frac{C^2}{\zeta^2} \quad \text{i.e., } \zeta = i \frac{\sqrt{2B}}{C\sqrt{T}} \sqrt{e} \quad (20)$$

$$\frac{dt}{dz} = \frac{i\sqrt{2B}}{2C\sqrt{T/2}\sqrt{e}} \quad (21)$$

$$\frac{dW}{dz} = \frac{dt}{dz} V \sin \beta \frac{T}{2} = \frac{iV \sin \beta \sqrt{T/2} \sqrt{B/2}}{\sqrt{e}} \quad (22)$$

$$m = iV \sin \beta \sqrt{T/2} \sqrt{B/2} \quad (23)$$

The m value here (last line) compared to the result from the baseline (the single-wing result) shows that they share the same formula, but the biplane expression is multiplied by an extra factor of $\sqrt{B}$. The factor B is a measure of how much the second wing blocks or shadows the vertical-flow edge velocity of the first wing ($B = 1$ for a huge gap with no blocking, and $B = 0.5$ for a tiny gap where each wing blocks half).

When Munk measures the longitudinal flow, he finds it does not exhibit this shadowing effect; its edge velocity is slightly increased rather than decreased by the presence of the other wing.

Overall, the types of flow considered are longitudinal flow, vertical flow, circulation flow, and countercirculation flow. For two wings, there are two trailing edges; Munk solves for the tuned values of P and Q that fix the singularity at the edges. To do this, he works out the edge velocities each flow type produces at the two edges, adds them up, and sets the total to zero at both edges. For this unstaggered, equal-wing case, he finds that $Q = 0$ (no countercirculation here), and P is given by:

$$P = TV(B \sin \beta + \sqrt{B} \sin \beta_0) \quad (24)$$

$$L = 2\pi V P \rho = 2\pi TV^2 \rho (B \sin \beta + \sqrt{B} \sin \beta_0) \quad (25)$$

Dividing and simplifying factors gives an expression for the lift coefficient:

$$C_L = 2\pi \sin \beta B + 2\pi \sin \beta_0 \sqrt{B} \quad (26)$$

He makes the approximation that $1 - \sqrt{B} \approx \frac{1}{2}(1 - B)$ for B close to 1, which is realistic in real-world gap-to-chord ratios. This explicitly shows the loss of lift compared to monoplanes through the factor B .

Finally, Munk computes a repulsion force since one wing's flow extends to the other and solves separately for circulation flow and vertical flow. He finds that the final force expression grows with the square of lift and is small for low angles of attack, but non-negligible at high lift.

2) *Unequal Wings, Varying Chords, and Decalage*: For upper and lower wings that have different amounts of camber, Munk uses the same effective curvature angle idea from before, except now there is one value per wing instead of the same amount for both. He finds that the circulation flow (responsible for tuning the P constant) depends on the average of the two curvature angles:

$$\frac{\beta_{01} + \beta_{02}}{2} \quad (27)$$

Since the two wings are not identical anymore, the counter-circulation flow actually has an effect now. This means the Q constant is not zero anymore due to the change in symmetry. The difference in curvature in the wings creates the need for this counter-circulation correction flow to smooth out the edge velocities:

$$Q = C \frac{T}{2} V \frac{\sin \beta_{01} - \sin \beta_{02}}{2} \sqrt{\frac{(1 - \lambda^2)(\lambda^2 - k'^2)}{B}} \quad (28)$$

$$= 2\lambda \sqrt{B} \frac{T}{2} V \frac{\sin \beta_{01} - \sin \beta_{02}}{2}$$

The Q constant redistributes lift, essentially determining how much of lift goes to each wing depending on curvature (more lift corresponds to the more curved wing).

a) *Chord*:: Munk states that for biplanes with two different chord lengths, the same transformation can be used which gives rise to similar integrals, but the math is redundant and incredibly complex. However, the results are not that significant or relevant, so Munk remains surface level. For chords with smaller differences in length, the change in behavior is irrelevant and they can be treated as wings of equal length.

b) *Decalage*:: Decalage is when the two wings are at different fixed angles relative to each other. The same transformation trick can't be used directly here since the wings aren't parallel. Instead, Munk uses an approximation method in which he calculates how much the flow around the lower wing influences the region where the upper wing sits if it weren't there. He finds the leaked velocity to be:

$$\frac{dW}{dz} = \left(1 - \frac{2z}{\sqrt{T^2 - 4z^2}}\right) V \sin \delta \quad (29)$$

$$\approx \tan \delta \left(1 - \frac{2y}{\sqrt{T^2 + 4y^2}}\right) \quad (30)$$

The bracketed factor d (shown in Fig. 11) demonstrates that the second wing only feels about 1/10 of the first wing's decalage angle for realistic gap-to-chord ratios [11]. Overall, for decalage, Munk finds that the total lift and moment stay nearly unchanged, but the amount of lift going to each respective wing changes significantly.

3) *Staggered Biplanes*: Stagger refers to shifting one wing forward or backward. The aerodynamics for stagger is much more complex because the symmetry is no longer clean. For

this scenario, both the P (circulation) and Q (counter-circulation) constants are needed. He uses the same transformation method, but now there is an extra constant a that controls how much stagger is introduced:

$$\frac{dz}{dt} = \frac{T}{2} C \left(\frac{t^2 - \lambda^2}{\sqrt{(1 - t^2)(t^2 - k'^2)}} + a \right) \quad (31)$$

For small amounts of stagger, the change in total lift and drag is very small. Munk finds the effect on the total to be proportional to the square of the stagger (accurate up to stagger = $\frac{1}{3}$ chord). Just like with decalage, the total lift stays the same, but how the lift splits between each wing changes significantly. In this case, it changes linearly with stagger (for smaller values).

Munk states that the usual arrangement has a greater lift for the rear wing at small angles of attack and greater lift for the front wing at greater angles of attack only. This flip occurs because of the interaction between the 2D stagger effect and the 3D induced drag effect: at low lift, 2D effects dominate, while at higher lift, 3D effects become more apparent.

4) *Aerodynamic Induction and Drag*: The induced drag for the biplane configuration is given by:

$$D_i = \frac{L^2}{k^2 b^2 q \pi} \quad (32)$$

where:

- b represents the greatest wing span,
- q represents the dynamic pressure ($q = \frac{1}{2} \rho V^2$), and
- k is the span factor (ranging from $k = 1.0$ for a tiny gap up to approximately $k = 1.21$ for larger gaps).

E. Forces Acting on an Airfoil

1) *Lift Force (General)*: Lift is the aerodynamic force generated perpendicular to the direction of the freestream flow, and it is the force primarily responsible for sustaining an aircraft in flight. Lift arises from a pressure differential between the upper and lower surfaces of an airfoil: as air accelerates over the curved upper surface, the local static pressure decreases relative to the lower surface, in accordance with Bernoulli's principle:

$$P + \frac{1}{2} \rho V^2 = \text{constant} \quad (33)$$

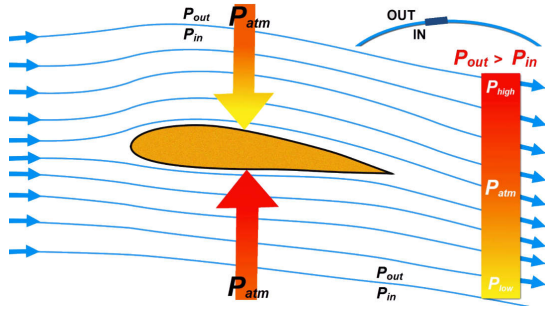
The net upward force produced by this pressure imbalance is given by [7]:

$$L = \frac{1}{2} \rho V^2 S C_L = q S C_L \quad (34)$$

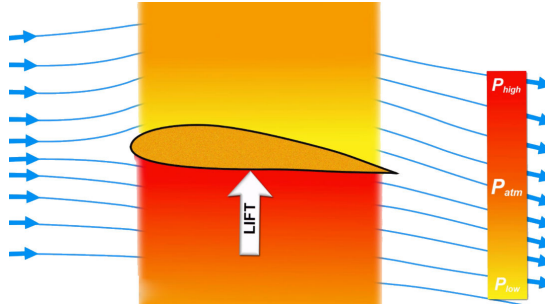
where L is the lift force, ρ is the freestream air density, V is the freestream velocity, S is the planform area, and C_L is the lift coefficient, which depends on airfoil shape, angle of attack, and Reynolds number. For a cambered airfoil, lift is generated even at zero angle of attack, since:

$$C_L = 2\pi(\alpha - \alpha_{L=0}) \quad (35)$$

where $\alpha_{L=0}$ is the zero-lift angle of attack determined by the airfoils camber [7].



(a) Bottom pressure should increase as airfoil is approached



(b) Pressure distribution around an airfoil

Fig. 10: Aerodynamic pressure analysis around an airfoil

2) *Lift Force (Biplanes)*: Despite having twice the amount of lift-generating surface, the airflow interference caused by the gap between the upper and lower wings reduces the overall generated lift, resulting in a net decrease in efficiency. Munk defines the lift coefficient as [2]:

$$C_L = 2\pi \sin \beta \quad (36)$$

He then applies biplane geometry to conformal mapping, which allows him to use similar circulation logic from a single wing to derive the lift coefficient for two wings. The final result is expressed as:

$$C_L = 2\pi \sin \beta \cdot B + 2\pi \sin \beta \sqrt{B} \quad (37)$$

The factor b (which is always less than 1, as shown in the reference table) represents the portion of lift lost by introducing a second wing. Munk also demonstrates that the $(1 - b)$ coefficient for biplanes is specifically applicable to the loss of lift associated with the angle of attack. Conversely, the loss of lift for cambered airfoils is found to be approximately $\frac{1}{2}(1 - b)$.

The intuitive reasoning for this stems from the flow behavior in cambered versus angled airfoils within biplane configurations. Angle-of-attack lift is driven by the shared airflow, causing one wing to shadow the other's contribution, whereas for camber, this shadowing effect is experienced primarily by one airfoil, reducing its overall impact. His results indicate an increased potential specifically for camber-based lift in

biplanes, despite the overall lift being reduced compared to single airfoils.

Kang et al. found that out of several parameters studied, gap and stagger have the most significant effects on the biplane configuration and are highly correlated with lift efficiency [11].

TABLE I.

Gap chord	B	B ₀	X	X B ₀	C	v	J	d	B ₁ (1+B)
0.0	1	1	1/4	0.250	0	0	0	0	1
57.7	0.997	0.999	0.250	0.250	-0.033	0.000	0.000	0.000	1.00
13.88	0.992	0.996	0.250	0.250	-0.238	0.000	0.000	0.000	1.00
3.76	0.982	0.981	0.250	0.250	-0.530	0.001	0.002	0.004	1.00
2.87	0.972	0.974	0.254	0.250	1.00	0.003	0.012	0.012	1.01
2.02	0.948	0.974	0.251	0.250	1.32	0.019	0.034	0.038	1.03
1.46	0.912	0.954	0.262	0.250	1.62	0.047	0.030	0.055	1.06
1.11	0.872	0.934	0.265	0.248	1.82	0.062	0.055	0.062	1.10
0.98	0.831	0.923	0.269	0.248	1.91	0.068	0.060	0.110	1.12
0.79	0.811	0.902	0.273	0.246	2.01	0.129	0.082	0.180	1.19
0.64	0.775	0.880	0.276	0.242	2.07	0.154	0.104	0.210	1.25
0.56	0.751	0.868	0.275	0.241	2.11	0.182	0.116	0.260	1.30
0.46	0.717	0.846	0.279	0.236	2.13	0.200	0.135	0.320	1.38
0.39	0.692	0.833	0.280	0.233	2.14	0.223	0.151	0.390	1.48

Fig. 11: Munk's results on biplane interference factors across gap-chord ratios

3) *Drag (General)*: Drag is the aerodynamic force that opposes an airfoil's motion through the air, acting parallel to the freestream direction. Analogous to lift, drag is calculated as:

$$D = \frac{1}{2} \rho V^2 S C_D = q S C_D \quad (38)$$

where C_D is the drag coefficient. Total drag on an airfoil is the sum of several components: skin friction drag, resulting from viscous shear stress within the boundary layer; pressure (form) drag, resulting from flow separation and the resulting pressure imbalance between the front and rear of the airfoil; and induced drag, resulting from the formation of trailing vortices. Because biplane configurations introduce additional surfaces and interference effects between the wings, drag (particularly induced drag) is a critical factor in evaluating biplane efficiency relative to monoplanes.

4) *Drag (Biplanes)*: Kang et al. demonstrate a correlation between the downwash angle and both gap and stagger. It is observed that the downwash angle increases with gap and stagger, and that it is directly proportional to the lift coefficient [11].

Munk defines the induced drag formula as [2]:

$$D_i = \frac{L^2}{k^2 b^2 q \pi} \quad (39)$$

Munk's findings demonstrate that $k = 1$ for a smaller gap and approximately $k = 1.21$ for a very large gap [2]. As follows, for that same total span, biplanes have lower induced drag than a monoplane because they induce less downwash on each other than a single wing. His drag equation and relation with k clearly highlight this relationship.

5) *Velocity and Dynamic Pressure*: For the purposes of this experiment, the freestream flow is treated as steady, incompressible, and low-speed, so the freestream density ρ can be treated as a constant rather than a variable that changes with velocity or pressure. This assumption holds well below Mach 0.3, consistent with the wind tunnel conditions used in this project. Under this assumption, dynamic pressure becomes a

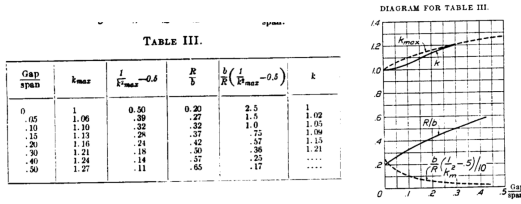


Fig. 12: Munk's results on gap and stagger effects on downwash.

function of velocity alone, which simplifies both the lift and drag equations introduced earlier [12].

Dynamic pressure, denoted q , is defined as [12]:

$$q = \frac{1}{2} \rho V^2 \quad (40)$$

and represents the kinetic energy per unit volume of the moving air. It is the quantity that converts the dimensionless lift and drag coefficients (C_L and C_D) into actual forces (L and D), as shown in the equations above. Dynamic pressure and static pressure are related through Bernoulli's equation:

$$P + \frac{1}{2} \rho V^2 = \text{constant} \quad (41)$$

which states that along a streamline in inviscid flow, the sum of static and dynamic pressure is conserved. Physically, this means that as velocity increases (as air accelerates over the upper surface of an airfoil), static pressure must decrease to keep the total constant. This is the mechanism responsible for the pressure differential described in, and it is the reason velocity and pressure measurements around the airfoil can be used interchangeably when calculating lift.

6) *Circulation*: Circulation, denoted Γ , is defined as the line integral of velocity around a closed curve enclosing the airfoil [10]:

$$\Gamma = - \oint_C \vec{V} \cdot d\vec{s} \quad (42)$$

Physically, Γ is a measure of how much flow curls around the airfoil. To clarify, a finite circulation does not mean the air is moving in circles, only that this integral evaluates to a nonzero value when taken around the body. Theoretically, circulation is the bridge between the airfoil's shape/AoA and the lift it produces.

The circulation formula is used to calculate lift within the wind tunnel. In practice, this inviscid relationship holds everywhere except within the boundary layer, the thin region of flow immediately adjacent to the airfoil surface where friction between the air and the surface impedes the flow. Outside this thin layer, the flow behaves essentially as predicted by inviscid theory.

7) *Viscid vs. Inviscid Flow*: A viscous (viscid) flow is one where friction, thermal conduction, and diffusion effects are present; an inviscid flow assumes none of these effects. True inviscid flow does not exist in nature, but for low-Reynolds-number flows like the ones in this experiment, it is an excellent

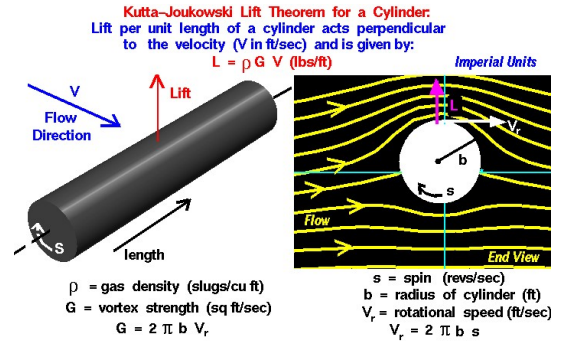


Fig. 13: Kutta Joukowski theorem for cylinder circulation.

approximation everywhere except the boundary layer. This assumption is adopted throughout: inviscid flow everywhere except within the boundary layer. This is what allows us to use circulation-based (rather than friction-based) methods to calculate lift, while still acknowledging that viscous effects in the boundary layer are the actual source of skin friction drag.

8) *Vorticity*: Vorticity, denoted ξ , is derived from circulation and is defined as the curl of the velocity field:

$$\xi = \nabla \times \vec{V} \quad (43)$$

Where circulation is a single value describing the net "swirl" around an entire closed curve, vorticity describes that same rotational tendency at a single point in the flow. By Stokes' theorem, circulation is simply the vorticity integrated over any open surface bounded by the curve, so a wing generating lift necessarily has a nonzero total vorticity in the flow around it.

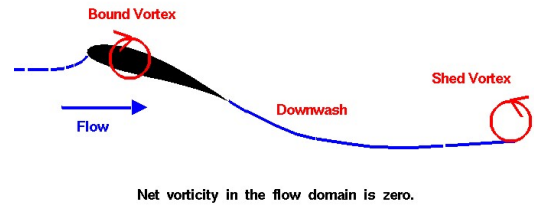


Fig. 14: The bound vortex on a Joukowski airfoil and the opposite-signed starting vortex shed into the wake

9) *Downwash*: One consequence of this vorticity is downwash: the trailing vortices shed from the wingtips induce a small downward velocity component in the vicinity of the wing itself. This tilts the local relative wind downward, reducing the effective AoA the airfoil feels and tilting the local lift vector backward, hence producing induced drag. As mentioned earlier with regard to biplanes, note downwash correlating with gap, stagger, and lift coefficient: two lifting surfaces in close proximity interact through each other's induced vorticity fields, changing the effective circulation (and therefore lift and

induced drag) each wing experiences compared to an isolate monoplane.

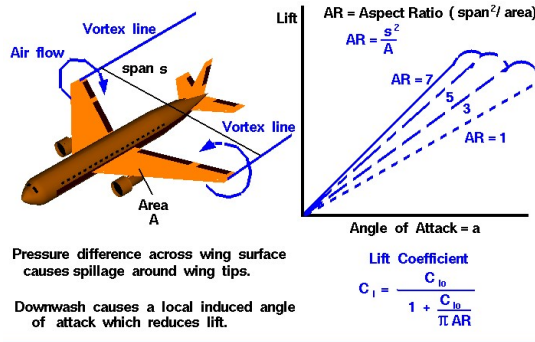


Fig. 15: The counter-rotating wingtip vortices and the resulting downwash region behind a finite wing

III. EXPERIMENTAL PROCEDURE

A. Introduction

The experiment was conducted to research the amount of lift produced by biplanes and the efficiency of biplanes. Prior to experimenting in wind tunnels, CFD simulations through ANSYS and XFLR5 were done on monoplane airfoils to determine properties of various airfoils on their own and the effects of changing different parameters on the airfoil: camber, max camber location and thickness. With the information gained from the CFD simulations, informed decisions about airfoils were made and 4 airfoils were chosen to be used for biplane experimentation in wind tunnels.

B. XFLR5 Simulations

XFLR5 is a virtual simulation software for airfoil analysis for airfoils at low Reynolds numbers. The simulation uses an estimation of the Reynolds Averaged Navier-Stokes equations to generate graphs and predict general trends for airfoils at varying angles of attack. Using this information, coefficient of lift vs. angle of attack and coefficient of lift vs. coefficient of drag graphs. For the simulations, the Reynolds number was set at 20,130, and a range of angles of attack from -20 to 30 with a α of .1. A selection of 16 airfoils was made and simulated in 4 groups of 4 airfoils. The airfoils were sorted by a control symmetrical group, airfoils with different cambers but everything else is the same, airfoils with different max camber location and airfoils with different maximum thicknesses. Grouping the airfoils this way made it possible to isolate one shape feature at a time to see its exact impact on performance. Testing each option across the full angle of attack range helped pinpoint key limits, like the exact point where lift reaches its maximum before dropping off. These initial software runs provided a quick and reliable baseline, showing which shapes produced the best balance of lift and stability before moving on to testing physical models in the wind tunnel.

In Fig. 16, four airfoils were tested with varying cambers, but all other variables were kept constant: NACA 0012, NACA

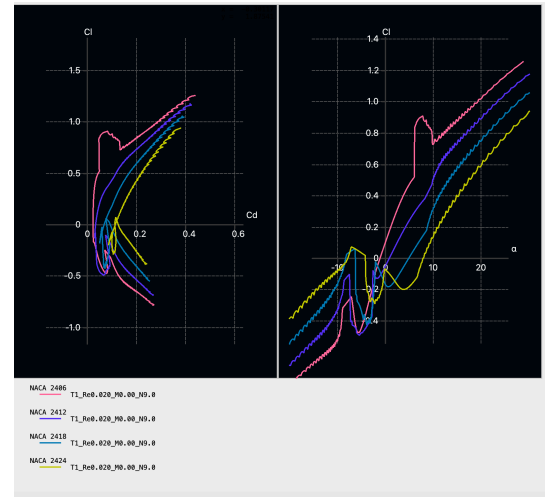


Fig. 16: XFLR5 simulation for varying cambers

2412, NACA 4412, NACA 6412. With this information, predictions about the critical angles and efficiency of the airfoil were able to be made. Using the information gained from these simulations, informed decisions were made on which were found most interesting and were further studied using ANSYS simulations.

C. ANSYS Fluent Flow

ANSYS fluent flow is a fluid dynamics simulation that was used to simulate the streamlines and velocities over the two dimensional airfoil [13]. Each of the six airfoils tested were run through ANSYS to obtain streamline and velocity contour graphics as well as coefficients of lift and drag at ten different angles of attack. This data was used to make informed decisions on which airfoils should be further tested in the wind tunnel. Following XFLR5 testing, six different airfoil geometries were imported into ANSYS from airfoil.com in the form of a tab delimited .txt file. Once imported, each geometry underwent a meshing process. This process forms an increasingly dense grid surrounding the airfoil, becoming more concentrated closer to the geometry. Each point on this grid is called a node while each space is called an element. ANSYS uses this grid to analyze the forces acting on each element and node and calculate an average of these individual values [13]. To simulate angle of attack in ANSYS without adjusting the orientation of the airfoil, each calculation had to be vectorized. ANSYS simulates velocity as flowing from the inlet, or front of the mesh. An airfoil generates lift in a perpendicular vector to the relative wind and drag in a tangential vector. If velocity, lift, and drag are not vectorized in the simulation, only the vertical portion of lift and horizontal portion of drag will be calculated, not the true vector. To counteract this, the cosine and sine of the AoA were entered into the X and Y portions of velocity and drag respectively, and the negative sine and positive cosine were entered into the X and Y portions of drag.

ANSYS analysis provided coefficients of lift and drag for each airfoil, as well as detailed graphics describing the airflow over the airfoil. However, these simulations were performed in two dimensions. Consequently, certain forces acting on a physical three dimensional airfoil were ignored in ANSYS. A three dimensional airfoil must consider not only air flowing over the top and bottom of the airfoil but also over the sides, or wingtips. This additional airflow generates downwash due to the formation of wingtip vortexes coming off the wing. Although a powerful and useful simulation, ANSYS in two dimensional format does not have the ability to calculate the additional force of downwash. This is the primary reason why further wind tunnel testing is necessary.

The Transition SST turbulence model, along with standard $k-\epsilon$ and $k-\omega$ models, is inaccurate here because it assumes a fully turbulent boundary layer from the leading edge, which fails at this Reynolds number.

A fresh initialization was used before every run so that previous runs would not influence the result for the active simulation. We found that if we did not initialize before calculating, the software would use the flow field from the previous simulation run and simply scale it up. For example, moving from $\alpha = 10^\circ$ to 15° with no initialization in between resulted in a flow field that looked identically colored, just larger for 15° . However, when initializing prior to calculating from 10° to 15° , we obtained two completely different contours that accurately represented the flow in each scenario.

The simulation finds a solution by guessing values close to the true value and getting progressively closer with each iteration. Initializing before calculating causes the guess to start from zero, whereas calculating without initializing causes the guess to start from the previous solution. A solution is "converged" when the guess homes in on a single value, whereas a solution is "unconverged" when it fluctuates between two or more values. This oscillation occurs when operating in a region where C_L and C_D are time-dependent.

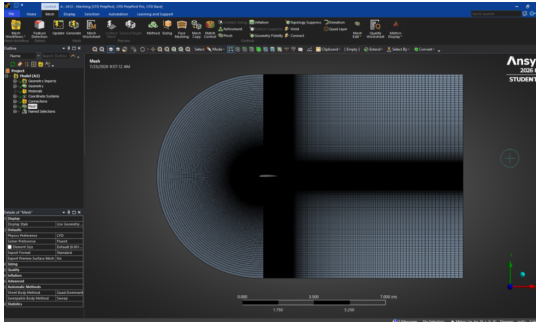


Fig. 17: Meshing for NACA 2412 Airfoil

D. Aspect Ratio / Finite Span Correction

Because the ANSYS simulations were two-dimensional while the wind-tunnel models had a finite span, a finite-span correction was applied to the simulated results before they were compared with the experimental data. The purpose of

this correction was to estimate the change caused specifically by transitioning from a 2D airfoil section to a 3D wing. This allowed the dimensional effect to be considered separately from differences caused by Reynolds number, turbulence, surface roughness, and other experimental conditions.

The physical models had a span of 150 mm and a chord of 30 mm. Since the airfoils had a rectangular planform, their aspect ratio was calculated as follows: $AR = \frac{b}{c} = 5$.

The 2D lift-curve slope, a_0 , was determined separately for each airfoil using the approximately linear pre-stall portion of its ANSYS C_l vs. α data. The slope obtained from ANSYS was used rather than the ideal thin-airfoil value of 2π , because the simulated slopes already reflected the low-Reynolds-number conditions used in this experiment. The corrected finite-span lift-curve slope was then calculated using

$$a_{3D} = \frac{a_0}{1 + \frac{a_0}{\pi AR e}} \quad (44)$$

where a_{3D} is the corrected three-dimensional lift-curve slope, a_0 is the slope obtained from ANSYS, AR is the aspect ratio, and e is the Oswald efficiency factor.

An Oswald efficiency factor of $e = 0.8$ is a commonly used approximate value for an unswept wing with a rectangular planform. This value was assumed rather than experimentally measured, and its effect on the correction is addressed further in the limitations section.

Airfoil	a_0 (rad)	α_{L0}	a_{3D} (rad)	Slope loss
0012	2.850	0.02°	2.323	18.5%
2212	3.554	-1.81°	2.770	22.0%
2412	2.496	-1.73°	2.082	16.6%
2612	1.894	-1.07°	1.646	13.1%

Fig. 18: Finite-span lift-curve slope corrections

The percentage loss in lift-curve slope was calculated using

$$\text{Slope Loss} = \left(1 - \frac{a_{3D}}{a_0}\right) \times 100\% \quad (45)$$

The correction predicted a reduction in lift-curve slope for all four airfoils. NACA 2212 had the greatest predicted reduction at 22.0%, while NACA 2612 had the smallest reduction at 13.1%. Using the corrected slope and the zero-lift angle obtained from the ANSYS data, the finite-span lift coefficient was approximated by

$$C_{L,3D}(\alpha) = a_{3D}(\alpha - \alpha_{L0}) \quad (46)$$

Angles were converted to radians when the lift-curve slopes were measured in rad^{-1} .

The 3D drag coefficient was estimated by adding the finite-span induced-drag contribution to the two-dimensional ANSYS drag coefficient:

$$C_{D,3D} = C_{d,2D} + \frac{C_{L,3D}^2}{\pi A Re} \quad (47)$$

The corrected C_L and C_D values were then compared with the wind-tunnel measurements. Since the ANSYS simulations and wind-tunnel experiments were conducted at approximately the same Reynolds number, applying this correction helped isolate the expected difference between a 2D airfoil and a physical finite-span wing. Any remaining difference between the corrected simulation and experimental result could then be examined in relation to other factors, including turbulence intensity, the 3D-printed surface, mounting interference, and wingtip leakage near the tunnel walls.

This correction was only applied to the approximately linear pre-stall region. It was not used to predict or adjust the stall angle because the lifting-line approximation does not represent the separated and time-dependent flow that occurs after stall.

E. Wind Tunnel Procedure

Using the CAD software OnShape, 4 different types of airfoils (NACA 0012, NACA 2212, NACA 2412, NACA 2612) were imported, modeled and 3D printed. Each type of airfoil was printed with a hole for a probe to be inserted at varying degrees, going from -5 to 20 angle of attack, increasing at an increment of 5. Angles of attack were manually calibrated using a precision protractor before testing to ensure that the alignment was accurate across all trials. The chord of the airfoil was 30mm and the length of the airfoil was 150mm [14].

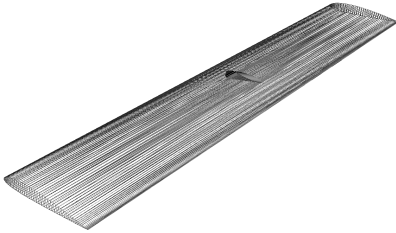


Fig. 19: NACA 2412 Airfoil CAD file with probe hole at 10 angle of attack

For the biplane configuration, a bracket was created to hold the two airfoils. The gap between the airfoils was 30 mm with no stagger.

The hole for the probe was placed at the quarter chord point in the center of the wing span of the airfoil, as seen in Fig. X. The quarter chord point is an approximation of the aerodynamic center found using the following equation:

$$x_{ac} = 0.25 + \frac{C_{m,0}}{C_{L,\alpha}} \quad (48)$$

Where x_{ac} is the location of the aerodynamic center as a percentage of the chord, $C_{m,0}$ is the pitching moment when the lift is 0, and $C_{L,\alpha}$ is the coefficient of lift per radian of angle of attack. Since the ratio between $C_{m,0}$ and $C_{L,\alpha}$ is very small, x_{ac} can be estimated as 25% of the chord. The aerodynamic center is significant since it is the point along the chord in which the pitching moment remains constant, regardless of the angle of attack of the airfoil [15]. This makes it a reliable reference point to place the probe in order to measure the lift and drag forces acting on the airfoil.

For the experimentation in the wind tunnel, a test matrix was created to determine the optimal configurations of the selected airfoils. In the test matrix, each airfoil was given a letter. The first letter represents the top airfoil, and the second represents the bottom airfoil. With this configuration, all possible combinations of given foils could be tested in the biplane configuration.

TABLE I: Test matrix for airfoil shape

	0012 (A)	2212 (B)	2412 (C)	2612 (D)
0012 (A)	AA	BA	CA	DA
2212 (B)	AB	BB	CB	DB
2412 (C)	AC	BC	CC	DC
2612 (D)	AD	BD	CD	DD

The control combination of airfoils was AA, representing two symmetrical airfoils, with the rest of the matrix including at least one cambered airfoil in each configuration. From the top performing airfoil combinations—greatest lift with the least drag—a second test matrix was created in order to test the optimal configurations at varying angles of attack and to determine the best angle of attack and the critical angle of attack for each airfoil combination. In the second matrix, the first number represents the angle of the top airfoil, and the second number represents the angle of the bottom airfoil. To ensure accurate force measurements during testing, the probe was again placed at the aerodynamic center of the airfoil, or quarter chord point. Expanding the test matrix across different angles of attack allowed for a full performance mapping, helping identify how biplane interaction affects stall onset compared to single airfoil performance.

Each airfoil configuration in each test matrix was placed into the wind tunnel.

The wind tunnel was designed so that in the forward direction of the airfoil, there was a large inlet that let air in, while aft of the airfoil there was a pump that pumped air out, creating an upstream velocity. In order to accelerate the air to a faster velocity, the cross-sectional area of the inlet narrowed in the contraction cone of the wind tunnel. Due to the Venturi effect, which states that cross-sectional area and velocity are inversely proportional, the velocity of the air increases and can be demonstrated in the following equation:

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 \quad (49)$$

TABLE II: Test matrix for airfoil angles

	-5°	0°	5°	10°	15°	20°
-5°	-5°, -5°	0°, -5°	5°, -5°	10°, -5°	15°, -5°	20°, -5°
0°	-5°, 0°	0°, 0°	5°, 0°	10°, 0°	15°, 0°	20°, 0°
5°	-5°, 5°	0°, 5°	5°, 5°	10°, 5°	15°, 5°	20°, 5°
10°	-5°, 10°	0°, 10°	5°, 10°	10°, 10°	15°, 10°	20°, 10°
15°	-5°, 15°	0°, 15°	5°, 15°	10°, 15°	15°, 15°	20°, 15°
20°	-5°, 20°	0°, 20°	5°, 20°	10°, 20°	15°, 20°	20°, 20°

where A is the cross-sectional area of the wind tunnel, v is the velocity of the air, and ρ is the density of air. Since the density of air is held constant ($\rho_1 = \rho_2$), it can be cancelled out of the equation and modeled as:

$$A_1 v_1 = A_2 v_2 \quad (50)$$

Through this effect, the wind tunnel is able to accelerate the wind to sufficient speeds.

By increasing the amount of air leaving through the pump, the velocity of the air can also be adjusted. By opening the valve more on the pump, more air is able to be drawn out, decreasing the pressure and therefore increasing the velocity of the air.

The wind tunnel contains an instrument that measures the dynamic pressure of the wind tunnel, which is the kinetic energy per unit volume of a moving liquid or gas. This pressure can be modeled in the following equation:

$$q = \frac{1}{2} \rho v^2 \quad (51)$$

where q is the dynamic pressure, ρ is the air density, and v is the velocity of the moving air. Using this equation, the velocity can be derived from the equation above to yield:

$$v = \sqrt{\frac{2q}{\rho}} \quad (52)$$

The dynamic pressure, and therefore the velocity, was held constant for all trials. With this technique, the velocity was calculated to be 10.07 m/s. This can be used to calculate the Reynolds number in the wind tunnel, which is 20,100.

Using the dynamic pressure along with the lift (L) which is obtained from experimental data in the wind tunnel and the surface area of the wing (S), the coefficient of lift (C_L) can be calculated with the following equation [7]:

$$C_L = \frac{L}{qS} \quad (53)$$

To compare the efficiency and drag performance, the coefficient of drag (C_d) is calculated using:

$$C_d = \frac{F_d}{qS} \quad (54)$$

where F_d is the measured drag force, q is dynamic pressure, and S is planform area. For individual baseline single-wing trials, the single-wing planform area ($S = b \cdot c$) was used. For both biplane test matrices, the combined planform area $S_{\text{total}} = 2S$ was substituted, as the force balance measures the net combined forces exerted on both wings as a single mounted system.

IV. RESULTS AND DISCUSSION

A. Overall Monoplane Results

For all 4 NACA airfoils tested, graphs of C_L vs α were created with the simulation results & experimental results plotted. Figures 27, 28, and 29 display the amount of over or under prediction for each α tested. From all 4 graphs, it is interesting to note that the experimental data tends to over-predict lift for α around 0 to 15 & under-predict for α from 0 to -15. Any over or underprediction of C_L is always within around 0.25. This recurring trend is likely due to the discrepancy between 2D and 3D prediction. Compared to each other, NACA 2212 dominates in the positive angle of attacks, but NACA 2612 has the highest overall C_L and tends to be superior after 15 degrees. The $C_{D,\text{min}}$ (minimum drag) and $C_{L,\text{max}}$ (stall point) can be found from the drag polar graph. The $C_{D,\text{min}}$ is around 0.04 for most airfoils except NACA 2212 which is 0.05. C_L is highest for NACA 2212 which is 0.9, though NACA 2412 and NACA 2612 are within 0.2 of this value. Finally, the Aerodynamic Efficiency plotted by angle of attack shows similar behavior for all 4 airfoils though the values vary. NACA 2212 has the highest lift to drag ratios, and varies by 2-3 from the others. The overall behavior increases and peaks at around 4-6 degrees, before decreasing.

B. Individual Monoplane Airfoil Results

1) *NACA 0012 Airfoil*: The ANSYS simulations found that for the airfoil, the C_L at $\alpha = 0^\circ$ was 0.00086 with a C_D of 0.0296. For $\alpha = 5^\circ$, the lift coefficient was $C_L = 0.2627$ and $C_D = 0.0562$. It was also observed that the airfoil experiences stall at some point between 0° and 5° angle of attack, as evidenced by the low-velocity recirculation zone (blue region) forming over the upper surface of the wing, indicating flow separation and the onset of a turbulent wake regime.

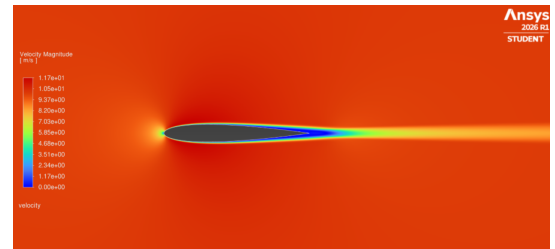


Fig. 20: Velocity contour of NACA 0012 at 0°

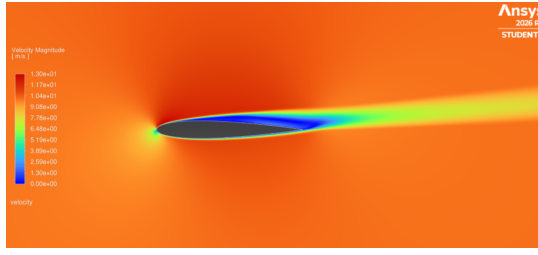


Fig. 21: Velocity contour of NACA 0012 at 5°

The results found in the figures differ from the experimental results obtained in the wind tunnel. The experimental results found that at a 0° angle of attack, the lift coefficient was $C_L = 0.0215$ and the drag coefficient was $C_D = 0.1577$. At 5°, the experimental values were $C_L = 0.3907$ and $C_D = 0.1720$. While the C_L and C_D data look different for the experimental case relative to the simulations, the rate of change of the lift coefficient with respect to α ($\frac{dC_L}{d\alpha}$) is consistent between both sets of results. This offset in magnitude is likely due to physical factors present in the wind tunnel that are not fully captured in the numerical models, such as surface roughness, mounting hardware interference, and ambient turbulence levels. However, the matching slope confirms that the underlying linear lift behavior holds true across both computational predictions and real-world testing.

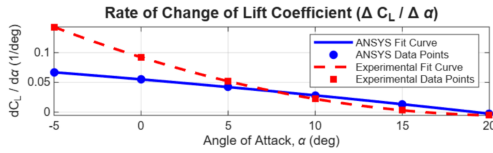


Fig. 22: Rate of Change of Lift Coefficient ($\Delta C_L / \Delta \alpha$)

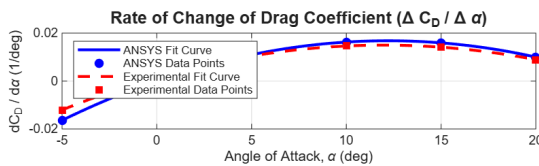


Fig. 23: Rate of Change of Drag Coefficient ($\Delta C_D / \Delta \alpha$)

The reason for the difference in experimental values compared to the simulated values may be a result of human error, issues with inconsistencies, surface irregularities, and the mounting probe interfering with the lift and drag force measurements. However, the graphs above do show that the simulations and the experimental data support each other since they follow the same trend, exhibiting very similar rates of change, especially for the C_D vs. C_L polar curves. The derivative trends confirm that the numerical simulations remain consistent with the experimental data.

2) *NACA 2212 Airfoil*: The simulations run on ANSYS and the experimental tests differed. The ANSYS simulation

predicted a stall around 10°; however, during wind tunnel experimentation, the airfoil did not stall within the testing range as predicted by ANSYS. In the simulation, C_L began to decrease after a 10° angle of attack, whereas C_L in the wind tunnel experimentation continued to increase and did not appear to stall.

The rate of change of C_L with respect to α ($\frac{dC_L}{d\alpha}$) is also inconsistent with the simulations. The average rate of change on the interval (15°, 20°), 0.0143, is greater than the average rate of change on the interval (10°, 15°), 0.108. In the simulations, the rate of change was negative and increasing, whereas the experimental rates of change were growing at an increasing rate.

TABLE III: Theoretical vs. Experimental Lift Coefficient (C_L) for NACA 2212 Airfoil

For 2212	C_L Theoretical	C_L Experimental
-5°	-0.464	-0.523
0°	-0.496	0.093
5°	0.385	0.505
10°	0.841	0.656
15°	0.782	0.710
20°	0.666	0.781

The discrepancy between the experimental and theoretical data could be a result of the formation of short laminar separation bubbles on the surface of the airfoil, increasing lift, that the simulations were unable to predict.

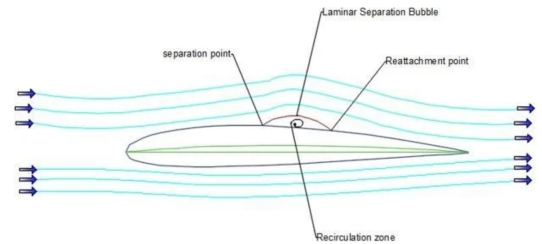


Fig. 24: Laminar Separation bubble diagram

A short laminar separation bubble occurs at low Reynolds numbers, where the flow temporarily separates from the airfoil surface along a portion of the chord before reattaching further downstream. As shown in Figure 24, the separation point marks where the streamlines disconnect from the surface to form the bubble, and they subsequently reattach at the reattachment point. The recirculating region beneath the separated shear layer consists of turbulent air, referred to as the circulation zone.

Because short laminar separation bubbles occupy a relatively small part of the chord, they can delay the onset of stall by forcing turbulent transition and reattaching the boundary layer to the airfoil surface. This mechanism provides a viable physical explanation for the experimentally observed C_L behavior, specifically why the physical airfoil did not experience stall at the same angle of attack predicted by theoretical C_L models.

However, laminar separation bubbles do not prevent stalling entirely. As the angle of attack increases, stall is typically triggered by one of two mechanisms: either the laminar separation bubble bursts, shedding an extensive turbulent wake that causes widespread flow separation, or the initial separation point moves far enough upstream that flow reattachment becomes impossible. Had the wind tunnel trials been conducted over extended run durations or higher resolution angle steps, this bubble bursting phenomenon might have been directly captured in the force measurements.

3) *NACA 2412 Airfoil*: Wind tunnel testing of the NACA 2412 deviated substantially from the results predicted in ANSYS. The experimentally obtained coefficient of lift at a -5° angle of attack was roughly 2.7 lower than the C_L solved from ANSYS. However, this was the only α where the simulation over-predicted lift. For all other α , the wind tunnel yielded a higher C_L than the simulation.

Wind tunnel testing also predicted a stall point earlier than the simulation. ANSYS analysis predicted a stall to occur slightly after 15° . On the other hand, the wind tunnel data placed the stall point at 10° , an inconsistency that was flagged and tested multiple times, yielding the same result. This discrepancy can be attributed to the airfoils geometry in relation to the wind tunnel walls. Air moving both beneath and along the sides of the airfoil is squeezed between the airfoil geometry and the walls of the wind tunnel, creating regions of higher pressure. In accordance with Bernoulli's principle, air in this high pressure region travels to areas of lower pressure, generating more lift than predicted using CFD.

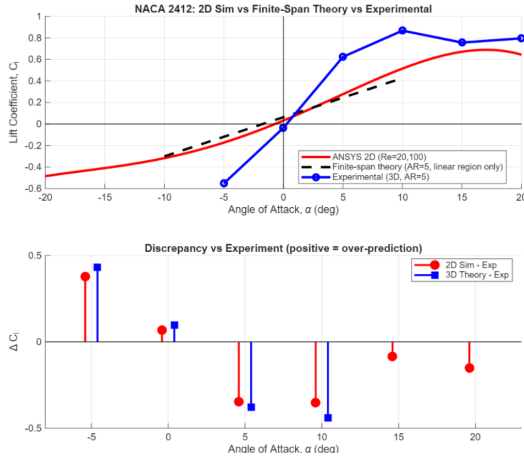


Fig. 25: C_L of simulation (red) vs. wind tunnel (blue)

An earlier stall point is also consistent with this logic. As air interacts with the wind tunnel walls, increased turbulence is generated within the test section. Furthermore, the airfoils boundary layer is disrupted by wall-bounded airflow interference.

Another phenomenon observed in the wind tunnel data was a slight recovery in lift past the stall point at $\alpha = 20^\circ$. Beyond stall, the separated turbulent flow exhibits unsteady, oscillating behavior over the upper surface of the airfoil. Consequently,

calculations of C_L and C_D become transient, yielding time-dependent values as turbulent structures convect across the wing surface.

4) *NACA 2612 Airfoil*: Physical wind tunnel testing of the NACA 2612 revealed results that consistently followed the primary trends predicted by ANSYS. The main discrepancy between CFD and experimental testing was the location of the predicted stall point. The NACA 2612 exhibited the most pronounced stall behavior among the individually tested airfoils: a sharp drop in performance that was not captured by ANSYS Fluent.

This difference is attributable in part to three-dimensional wing effects. Two-dimensional CFD simulations cannot account for 3D flow phenomena that influence finite wing performance. Although the airfoils were positioned in the wind tunnel to minimize effects such as downwash and wingtip vortices, their influence was not entirely eliminated.

The NACA 2612 airfoil achieved its maximum aerodynamic efficiency (C_L/C_D) at higher angles of attack, peaking near $\alpha = 10^\circ$. Additionally, this profile exhibited a post-stall lift behavior similar to the slight lift increase observed in the NACA 2412.

C. Biplane Wind Tunnel Results

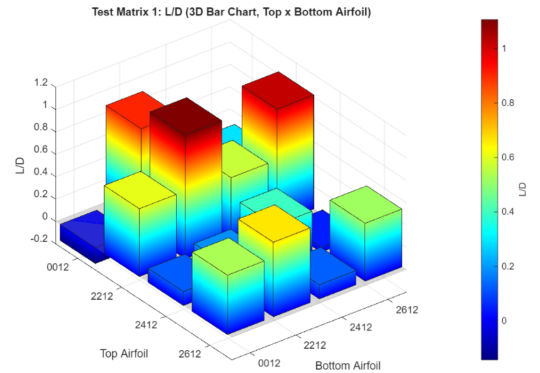


Fig. 26: Test matrix 1: L/D (3D Bar Chart, Top x Bottom Airfoil)

For all four NACA airfoils selected, a test of every possible combination was performed, yielding 16 distinct dual-airfoil configurations. The results are demonstrated through the heatmap above. From these tests, the combination with the highest lift-to-drag ratio featured the NACA 2212 on top and the NACA 2212 on the bottom, achieving an L/D ratio of 1.1064.

Most of the drag coefficient values lie within a narrow range between 0.07 and 0.1, as shown by Figure 28. The lift range, tracked by circulation, is significantly larger and served as the primary tuning factor for the overall lift-to-drag ratio. Figure 27 and Figure 29 demonstrate the larger spread of C_L compared to C_D , while also highlighting how circulation closely tracks C_L . Furthermore, Figure 29 exhibits a very similar shape to Figure 27, illustrating that the efficiency ratio is far more dependent on lift variations than drag variations.

The NACA 2212 by NACA 2212 combination optimizes camber without inducing excessive aerodynamic interference between the top and bottom wings. While a combination with a higher maximum camber location might have increased lift independently, it would have generated greater flow interference from the lower wing, resulting in an overall net reduction in total lift.

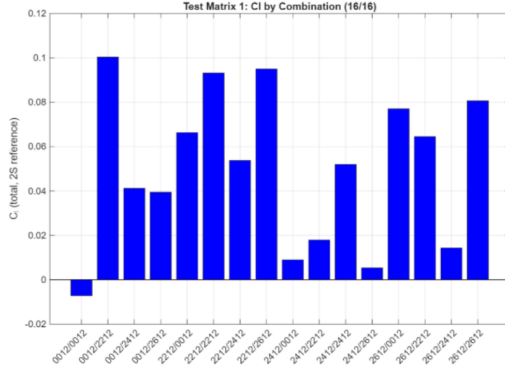


Fig. 27: Test matrix 1: C_L by combination (16/16)

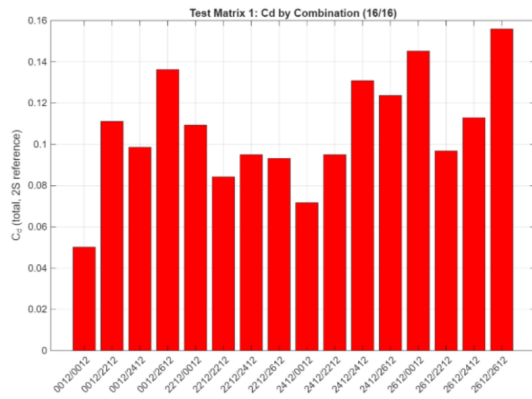


Fig. 28: Test matrix 1: C_D by combination (16/16)

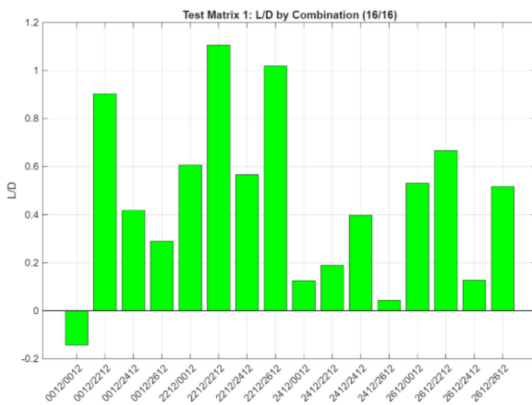


Fig. 29: Test matrix 1: L/D by combination (16/16)

It can also be seen through the specialized equation for lift (26) that 2212 x 2212 results in the greatest C_L compared to

the other combinations. Though this doesn't account for drag which is essential for determining aerodynamic efficiency, it can be used to demonstrate the reduction in lift for different max camber locations in a computational way which lines up with the experimental results.

The next best combination had the 2212 on the top and the 2412 on the bottom with a value of 1.0192. Similarly, this is due to the optimization of camber with airflow interference. The worst performing combination of airfoils had a 0012 on the top and bottom, producing a ratio of -0.143. As 0012 is the symmetric airfoil, a zero angle of attack does not result in any lift due to the lack of camber. Drag and airflow interference are still produced which further lower the aerodynamic efficiency, making it the worst combination.

All combinations of airfoils except for the two best performing ones had a lift to drag ratio of less than one, meaning that the drag force is greater than the lift produced by the wing. Due to having two wings, drag is significantly increased. Optimizing the ratio requires a combination that not only considers maximizing lift and reducing drag of a single airfoil, but how it influences the other. Therefore due to the sensitivity of airflow interference, it is reasonable that a large portion of the combinations had a negative lift-to-drag ratio. With 2212 by 2212 determined as the best configuration, the second test matrix was completed.

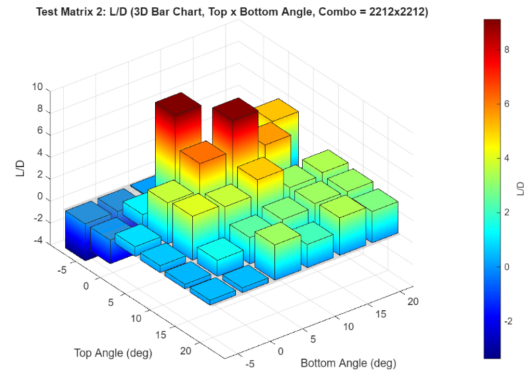


Fig. 30: Test matrix 2: L/D (3D Bar Chart, Top x Bottom Angle, 2212 x 2212)

The results of the second matrix are clearly shown in figure 26. In the second test matrix, 36 different combinations of varying angles were tested, with the top and bottom airfoil being at different angles using the 2212 x 2212 combination. From this, the best lift to drag ratio was found. This was done with the top airfoil at 0° and the bottom angle of attack at 5° which resulted in a ratio of 9.12, which was retested 4 times to confirm [16]. The other well performing combinations were 5 on top and 10 on bottom (8.98) as well as 5 on top and 5 on bottom (6.19).

It is interesting to note that the three worst performing decalage combinations did not particularly produce a large amount of drag (top -5° and bottom -5° , top 0° and bottom -5° , and top -5° and bottom 0°). The main source of this

inefficiency came from the amount of lift force. Essentially, a negative angle of attack significantly increases the interference of flow between the two airfoils. Furthermore, as Munk mentions, the average of the two angles can be used to determine the C_L for combinations with decalage. Plugging in values into his equation demonstrates the same relation with decreased lift, illustrating the consistency experimentally and computationally. Conversely, with increased angle of attack, Munk's equation should result in higher C_L values. This holds true, but the C_D also increases which ends up reducing overall efficiency. The best combination (5° and 0°) lies in optimizing both, in which decalage is increased enough to produce the greatest lift without inducing significant drag.

1) *Further Discussion on 2212-2212 at $0^\circ \cdot 5^\circ$ and $5^\circ \cdot 10^\circ$:* The biplanes at $0^\circ \cdot 5^\circ$ and $5^\circ \cdot 10^\circ$ performed significantly better than the other combination of angles. They had a lift to drag ratio of 9.12 and 8.98 respectively. These biplanes both shared a converging geometry on the channel between the airfoils. This was initially hypothesized that it behaves analogously to a converging nozzle, such as on a rocket. The wind going through the gap is accelerated by Bernoulli's principle, creating some thrust [17].

While a rocket nozzle can be used as an analogy for this effect, a passize, unpowered wing cannot generate a net thrust [17]. So instead, the local acceleration of the air most likely contributes to a reshaping of the pressure distribution on the wing surfaces. This creates a more concentrated Venturi effect in the region.

Both airfoils above showed reduced C_D due to this acceleration of air through the gap of the airfoil. This, along with a high C_L value, increases the overall lift to drag ratio relative to the rest of the biplanes.

This increase in overall efficiency can help to drastically reduce the fuel mass for a given range. Since the wing is more efficient, the aircraft requires less fuel, making it more efficient. This relationship can be modeled by the Breguet Range Equation:

$$R \propto \frac{L}{D} \ln \left(\frac{W_i}{W_f} \right) \quad (55)$$

This creates a compounding relationship, as the required fuel and efficiency of the aircraft decrease significantly, as modeled by the logarithmic function. This concept is analogous to the famous Tsiolkovsky rocket equation used in rocketry, which also states that as the mass of the vehicle increases, the propellant mass grows sharply because the vehicle must hold its own propellant.

While this effect is less significant in aircraft, even a small increase in lift to drag ratio can lead to large benefits and increased payload over longer ranges in biplanes opposed to monoplanes.

2) *Stall-Order Asymmetry:* With the data from the second test matrix, an asymmetrical stall order relationship between the two airfoils can be seen.

The top airfoil stalling has a greater effect on the overall lift than the bottom airfoil stalling. When the bottom airfoil stalls

before the top airfoil, the resulting loss of lift significantly less and depends more on the angle of the top airfoil.

All of the best performing biplanes had a greater angle of attack on the bottom airfoils relative to the top airfoil. Meanwhile, all the worst performing airfoils had the top airfoil with the greater angle of attack.

The bottom airfoil having a larger angle of attack than the top can help to fix an aircraft in a stall. By one wing stalling before another, it causes a decrease in lift, and allows the aircraft to return to a stable angle of attack since the other airfoil would still be producing lift.

This is conceptually similar to the canard wing configuration on airfoils. Canard configured aircraft have a large main wing and smaller forward wings with a different camber to help prevent stalling. This similar idea can be seen with the top and bottom biplane configuration tested.

With this information, it does motivate the idea of a biplane with variable geometry and an extending wing. For example, it would be more efficient if a plane would be able to deploy a second wing for low-speed and during phases needing high lift, such as takeoff and landing. Using this data, an aircraft with these characteristics could be created.

This characteristic is also consistent with the channel-flow explanation above. With the bottom airfoil being at a higher angle, the overall airfoil performance is increased due to the asymmetry of the stalls and the channel flow effect improving the efficiency of the biplane.

D. Limitations

Throughout testing, certain variables were assigned values instead of being experimentally determined. A turbulence intensity (T.I.) of 0.5% was assumed during analysis. T.I. is a value used to determine the strength of the turbulent airflow over the airfoil. The exact T.I. of a wind tunnel can be acquired using hot wire calibration, where a hot wire probe is inserted into a wind tunnel to collect data on airspeed velocity over time [16]. When conducting ANSYS simulations, 0.5% was used for T.I. instead of the actual T.I. for the wind tunnel used in experimentation. This factor is partially responsible for the early stall point observed in certain airfoil geometries in the wind tunnel.

According to thin airfoil theory, any cambered airfoil should generate lift at zero α . However, this concept is especially susceptible to T.I. Consequently, testing of the NACA 2412 and NACA 2612—both cambered airfoils—exhibited slightly negative values for lift at zero degrees. While it is true that this is transient, or time sensitive data, these data points were retested and experimentally verified. Another source for the discrepancy between theoretical and experimental results could be a result of wall proximity effects. The airfoils were intentionally placed close to the wind tunnel walls to minimize the effect of wingtip vortices forming and downwash. However, there was a .5mm gap between the wingtip and the wall. The high pressure air from below the wing moves around the wingtip to the low pressure region above it. This is the same process that wingtip vortices form on free wings. Due

to the same amount of air being forced into a smaller gap, the air is accelerated, by Bernoulli's principle, which lowers the pressure, pulling more air over the wingtip, decreasing the lift produced. While the downwash effect was likely still reduced due to the airfoils proximity to the edge of the wind tunnel, without a perfect seal, wingtip leakage would still occur, affecting the results. This leakage should not be mistaken for wind tunnel wall interference, however. There is no physical path connecting the airfoil to the wall. The true behavior of a wing with the aspect ratio of 5 might not be represented by these results and end up somewhere in between a model that is fully bound to the walls and a wing in free-air due to the leakage around the wingtips

Furthermore, Oswald efficiency number, or e , was assumed to be 0.8 for the finite span correction. Since e affects both the corrected lift-curve slope and induced-drag calculation, any difference between the assumed and actual value would alter the magnitude of the correction. Additionally, the correction assumes a wing in free air, while the models were positioned close to the wind-tunnel walls with a small wingtip gap. Therefore, wall proximity and tip leakage effects may cause the actual efficiency factor to differ from the assumed value. More extensive testing would be required to determine a wing-specific value of e .

E. Future Research Plans

Possible future research could expand the scope of this current research by experimenting with a wider range of biplane metrics, including biplane gap, stagger, and decalage. In this study, the airfoil gap was fixed and no stagger was used, so the performance that was observed cannot be generalized to other biplane arrangements. A more comprehensive study would help determine whether the high efficiency of NACA 2212 x NACA 2212 at 0 degrees upper wing and 5 degrees lower wing would remain optimal if biplane gap and stagger were to change. It would be valuable to test a larger variety of airfoils with factors other than max-camber location changed and including multiple iterations per sweep along with smaller angle increments to truly optimize the configuration. Furthermore, the forces acting on the upper and lower wings should be measured separately. The current force measurements measured their combined lift and drag of the entire biplane system, making it hard to determine how decalage redistributed the aerodynamic loading between the two wings. A future study could also look into testing the aerodynamics of biplane configurations in supersonic speeds, possibly using the learned insights to implement the theoretical busman biplane or engineering a canard biplane structure.

V. CONCLUSIONS

This study evaluated four NACA airfoils —0012, 2212, 2412, and 2612—in monoplane and biplane configurations through computational analysis and wind-tunnel testing. The first biplane matrix compared all 16 upper- and lower-airfoil combinations and identified NACA 2212 x NACA 2212 as the most efficient pairing, with $L/D = 1.11$. A second

matrix then test 36 angle of attack configurations of this pairing. The highest measured efficiency was $L/D = 9.12$, produced with the upper wing at 0° and the lower wing at 5° , and was reproduced across four trials. The best-performing configurations generally paired a small positive upper-wing angle with an equal or greater lower-wing angle. Because drag varied over a relatively narrow range while lift changed substantially, the differences in L/D were primarily driven by changes in lift generation.

A central methodological finding was that the disagreement between 2D simulation and 3D simulation wind-tunnel data should not be attributed to a single effect. Low-Reynolds-number behavior first changes the lift slope, separation behavior, and profile drag of the airfoil section itself. Converting that section into a finite-span wing then introduces an additional reduction in lift-curve slope and an increase in induced drag. Applying the finite-span correction therefore helped distinguish the effects of Reynolds number from those caused by the transition from a 2D airfoil to a 3D wing.

The results also suggest that biplane performance depends on the interaction between the two wings rather than simply the combined behavior of two isolated airfoils. The observed sensitivity to decalage may be useful in future variable-geometry designs, where a second wing could be deployed during low-speed, high-lift phases and repositioned or retracted during cruise. The apparent stall-order asymmetry also provides a possible comparison to canard configurations, although pitching moment and aircraft stability were not measured in this study. Similarly, the Breguet range relationship indicated that improvements in L/D could increase range or reduce fuel requirements, but full-aircraft performance was outside the scope of the experiment.

Future work should repeat the remaining decalage trials, measure turbulence intensity and wing-specific efficiency directly, and examine the forces acting on each wing separately. Three-dimensional CFD would also help resolve the flow interaction within the gap, particularly, the small clearance between the wingtips and tunnel walls. By combining low-Reynolds-number simulation, finite-span correction, and systematic testing of both airfoil pairing and decalage, this study provides a rare experimental baseline for modern biplane optimization. The results show how the second wings geometry and incidence reshape the coupled flow field and can substantially alter overall efficiency. This establishes wing-to-wing interaction as a meaningful design variable and supports renewed investigation of biplanes for low-speed, high-lift, and variable-geometry applications.

VI. ACKNOWLEDGEMENTS

The authors of this paper are sincerely grateful for the many contributors who dedicated themselves to this research paper's success and would like to acknowledge the following: Project Mentor Krishiv Patel for his admirable expertise, attention, and guidance; Heather Fenton for her commitment and support as a Residential Teaching Assistant; Research Coordinator Vinay Menon for organizing this invaluable research opportunity;

Dr. Edward DeMauro for his professional insight; Dean Jean Patrick Antoine, the Director of NJ GSET, for his dedication to the Governors School and providing the opportunity, resources, and support for scholars to expand their learning. The writers of this paper kindly acknowledge the Governors School of New Jersey Program in Engineering and Technology (GSET), Rutgers University, Rutgers School of Engineering, the State of New Jersey, former Governors School scholars and parents, the Governors School Alumni Association, Lockheed Martin and other program sponsors. We thank Melvin Qiu, for his generosity and 3D printer in unprecedented circumstances, and finally, all of the scholars who took part in the 2026 GSET cohort, for their camaraderie, curiosity, and memories.

REFERENCES

- [1] C. T. Sun and C. Lu, *Mechanics of Aircraft Structures*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2019.
- [2] M. M. Munk, "General biplane theory," National Advisory Committee for Aeronautics (NACA), Tech. Rep. NACA Report 151, 1922. [Online]. Available: <https://scispace.com/pdf/general-biplane-theory-59pd2d0t3p.pdf>
- [3] J. D. Anderson, Jr., *Fundamentals of Aerodynamics*, 5th ed. New York, NY: McGraw-Hill, 2011.
- [4] Embry-Riddle Aeronautical University, "Airfoil characteristics," in *Introduction to Aerospace Flight Vehicles*. Eagle Publications, 2021. [Online]. Available: <https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/airfoil-characteristics/>
- [5] NASA Glenn Research Center. (2021) Inclined airfoil. Accessed: 2026. [Online]. Available: <https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/incline.html>
- [6] SKYbrary. (2022) Angle of attack (AoA). Accessed: 2026. [Online]. Available: <https://skybrary.aero/articles/angle-attack-aoa>
- [7] NASA Glenn Research Center. (2021) The lift equation. Accessed: 2026. [Online]. Available: <https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/lifteq.html>
- [8] Fossil Consulting Services. (2020) Power plant fundamentals: Laminar flow. Accessed: 2026. [Online]. Available: <https://www.fossilconsulting.com/blog/power-plant-fundamentals/laminar-flow/>
- [9] Federal Aviation Administration, *Airplane Flying Handbook*. Washington, DC: Federal Aviation Administration, U.S. Department of Transportation, 2023, no. FAA-H-8083-3C.
- [10] C. Bai and Z. Wu, "Generalized Kutta-Joukowski theorem for multi-vortex and multi-airfoil flow (a lumped vortex model)," *Chinese Journal of Aeronautics*, vol. 26, no. 4, pp. 856–866, 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1000936113001581>
- [11] H. Kang, "Gap and stagger effects on the aerodynamic performance and the wake behind a biplane with endplates," Doctoral dissertation, University of Dayton, Dayton, OH, 2008. [Online]. Available: https://ecommons.udayton.edu/graduate_theses/3623
- [12] NASA Glenn Research Center. (2021) Dynamic pressure. Accessed: 2026. [Online]. Available: <https://www.grc.nasa.gov/www/BGH/dynpress.html>
- [13] K. L. Lawrence, *ANSYS Tutorial Release 2023: Structural and Thermal Analysis Using the ANSYS Mechanical APDL Release 2023 Environment*. Mission, KS: SDC Publications, 2023.
- [14] NASA Glenn Research Center. (2021) Wind tunnel nozzles. Accessed: 2026. [Online]. Available: <https://www.grc.nasa.gov/www/k-12/airplane/tunnoz.html>
- [15] R. C. Nelson, *Flight Stability and Automatic Control*, 2nd ed. Boston, MA: WCB/McGraw-Hill, 1998.
- [16] R. S. Figliola and D. E. Beasley, *Theory and Design for Mechanical Measurements*, 7th ed. Hoboken, NJ: John Wiley & Sons, 2020.
- [17] D. K. Huzel and D. H. Huang, *Modern Engineering for Design of Liquid Propellant Rocket Engines*, ser. Progress in Astronautics and Aeronautics. Washington, DC: American Institute of Aeronautics and Astronautics, 1992, vol. 147.